Phishing Website Anomaly Detection: Reconstruction Scores, Feature Selection, and Operational Trade-offs

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GitHub repository: https://github.com/v1852465cs15/HW2-Anomaly-Detection

Abstract

Phishing alerts must protect users without overwhelming analysts. On 11,055 tabular records, a normal-only categorical AE achieves five-seed AP .940, exceeding heuristic/Isolation Forest baselines but trailing supervised XGBoost (.998). Top-9 MI features retain close performance; redundancy conclusions depend on the cutoff. Training contamination of 10% reduces AP from .939 to .709. Additive explanations and six failure cases expose legitimate-like phishing and rare benign profiles. At 1% prevalence, FPR constraints reduce projected false alarms from 701 to 99 per 10,000 scans at lower recall. Duplicate vectors and untested distribution shift limit deployment claims.

1 Introduction

We study **Tabular** phishing detection with 30 categorical indicators, without URLs/HTML. A feature-wise softmax AE models code combinations without numeric-distance assumptions. Legitimate `train_00000` has (SSL, anchor)=$(1, 1)$; phishing `train_00002` has $(-1, -1)$: coded examples, not live inspections.

Security analysts review alerts, warn users, and block confirmed threats. FPs waste review capacity/disrupt visits; FNs expose users to credential theft.

Research question: Can normal-only reconstruction detect phishing while preserving performance with nine features and controlling false-alarm workload?

Our contributions are:

(1) a shared-split three-tier comparison with explicit supervision differences.

(2) training-only MI/redundancy evidence and controlled feature, contamination, and scoring ablations.

(3) score-level explanations, concrete FP/FN cases, and prevalence-adjusted workload estimates.

2 Related Work

Our category is **Tabular**; the following studies differ in data, method, supervision, and outcomes.

Hannousse and Yahiouche (2021): 87 URL/content/service features, supervised Random Forest, accuracy 96.61% hybrid/96.83% after selection. We instead test normal-only fitting and redundancy on 30 tabular codes.

Bu and Cho (2021): 222,541 URLs, three corpora, character-level convolutional AE plus supervised auxiliary CNN; reported sensitivity gain 3.98%. We reconstruct tabular codes with additive explanations.

Sahingoz et al. (2019): 73,575 URLs, supervised Random Forest/NLP, accuracy 97.98%. We compare supervision and alert costs; cross-dataset scores are contextual.

Tier assumptions: -1 counts reflect risk (Tier 1 floor); normal combinations reconstruct well or anomalies isolate easily (Tier 2 AE/Isolation Forest; Liu et al. (2008)); labeled phishing generalizes (Tier 3 XGBoost; Chen and Guestrin (2016)).

3 Task Definition & Data

We use UCI Phishing Websites: 11,055 records, 30 categorical features, sourced from PhishTank, MillerSmiles, and Google search operators (Mohammad and McCluskey, 2012). Inherited labels define legitimate ($Result = 1, y = 0$) versus phishing ($Result = -1, y = 1$). Anomaly means phishing, not rarity: 4,898 (44.31%) phishing and 6,157 (55.69%) legitimate, approximately $1 : 1.26$.

Always-normal achieves 55.69% accuracy but misses every phishing site. This moderate imbalance and deployment-prevalence mismatch motivate AP, recall, FPR, and workload projections at

Partition	Total	Legitimate	Phishing
Training	7,738	4,310	3,428
Validation	1,658	923	735
Test	1,659	924	735

Table 1: Our stratified split; one-class models fit the legitimate training subset only.

1% prevalence rather than accuracy.

The 70/15/15 split retains 5,206 extra duplicate labeled rows, 5,270 extra feature vectors, and 64 conflicting-label groups, suggesting representation ambiguity or annotation noise. No features are missing/constant. URL, HTML, form, and reputation indicators are categorical; zero is not missingness.

4 Exploratory Feature Analysis

Mutual Information Ranking. Higher MI indicates more information about legitimate versus phishing labels. We compute MI in nats on 7,738 training records only; this label-assisted selection is separate from normal-only AE fitting. Figure 1(a) ranks all features. The Top-9 are SSLfinal_State, URL_of_Anchor, Prefix_Suffix, web_traffic, having_Sub_Domain, Links_in_tags, Request_URL, SFH, and Domain_registeration_length.

The exploratory cutoff follows MI's 3.43-fold drop from rank 9 (.02461) to 10 (.00717), motivating 70% feature reduction with Top-5/15 controls. It is not uniquely optimal; lower-MI features may remain useful.

Normal versus phishing. Figure 1(c): SSL code 1 is 89.12% legitimate; -1 is 85.74% phishing. In (d), anchor code -1 is 99.00% phishing; 1 is 94.15% legitimate. These within-code class shares show overlap; neither feature determines the label.

Feature Association. Higher Cramer's V in Figure 1(b) suggests overlapping information. We use training-only Pearson counts, $V = \sqrt{\chi^2/[n\min(r-1, c-1)]}$, without Yates correction. The exploratory cutoff $V \geq .80$ was not preregistered.

Features	Pairs	$V \geq .50$	$V \geq .80$	Max V
All 30	435	25	4	.939
Top 9	36	1	0	.606

Table 2: Training-only pairwise redundancy sensitivity ($n = 7,738$); diagonal and mirrored pairs are excluded.

Four pairs reach $V \geq .80$: Favicon/popUpWidnow (.9394), URL shortener/double-slash redirect (.8405), Favicon/port (.8027), and port/email submission (.8008); Figure 1(b) gives feature codes. Top-9's maximum, registration length/Request_URL (.6057), exceeds .50/.60. Redundancy is cutoff-dependent; pairwise V cannot establish conditional complementarity.

H1: Selecting the top 9 features based on Mutual Information can achieve comparable anomaly detection performance to using all 30 features.

H2: The top 9 features provide sufficient complementary information for phishing website detection without highly redundant features.

5 Method

5.1 Categorical reconstruction

Three-position one-hot encoding gives 90 inputs without fitted vocabularies/scalers. The AE is $90 \to 128 \to 64 \to 8 \to 64 \to 128 \to 90$, with feature-wise softmaxes (Figure 2); categorical CE fits only 4,310 legitimate training records.

$$s(x) = \sum_{j=1}^{30} s_j(x), \qquad s_j(x) = -\log \widehat{p}_j(x_j \mid e(x)). \tag{1}$$

Larger scores indicate reconstruction surprise; additive feature terms explain the total in nats.

5.2 Threshold decision and error costs

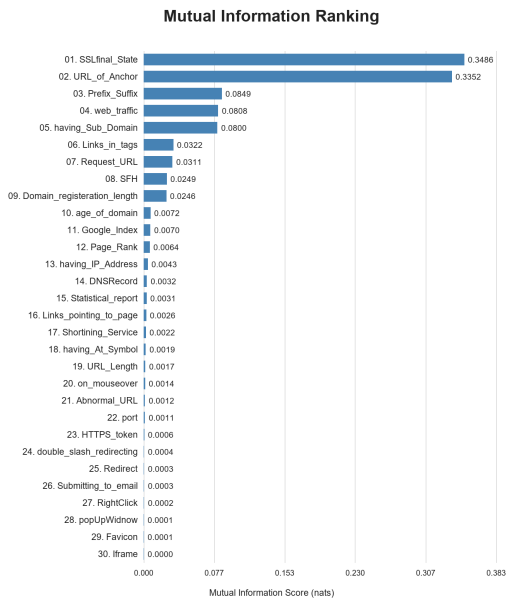
Alert when $s(x) \geq \tau$. Validation-F1 optima are 6.696, 9.230, 8.804, 10.531, 8.691 nats across five seeds (mean 8.79). F1 balances precision/recall but specifies no monetary FP/FN costs and may exceed review capacity at low prevalence.

Alternatively, constrain legitimate-validation FPR to 1%: suppress costly false warnings/reviews while accepting more FNs. Test FPR is measured separately, not guaranteed. Neither threshold rule uses test labels.

5.3 Comparison methods

Tier 1: always-normal and count of -1 codes (validation threshold 7), intentionally naive because signs have feature-specific meanings. Normal-only Isolation Forest: 300 trees, 1,024 max samples, .5 max features. Supervised XGBoost: both classes, 500 trees, depth 9, learning rate .05; phishing probability is its score, with a label advantage.

(a) MI ranking



(b) Feature association

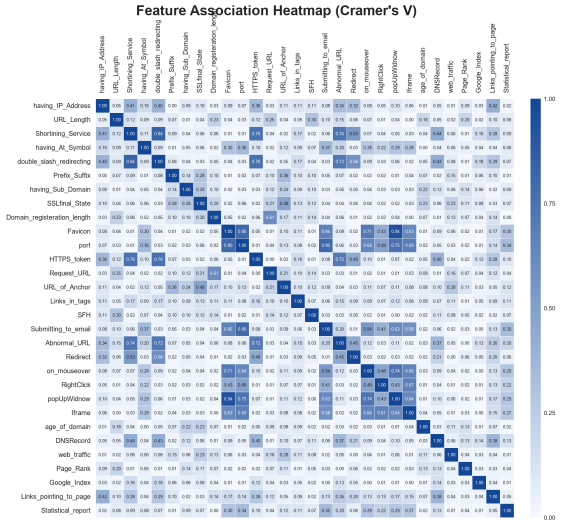
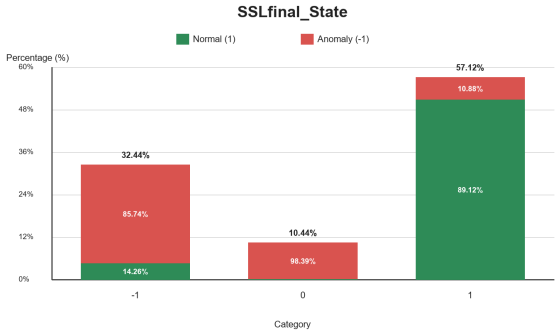


Figure 1: Training-only feature analysis. (a) MI ranking and (b) Cramer's V heatmap with feature names and numerical annotations. Class-conditional evidence is continued in (c)–(d).

(c) SSL-state distribution



(d) Anchor-code distribution

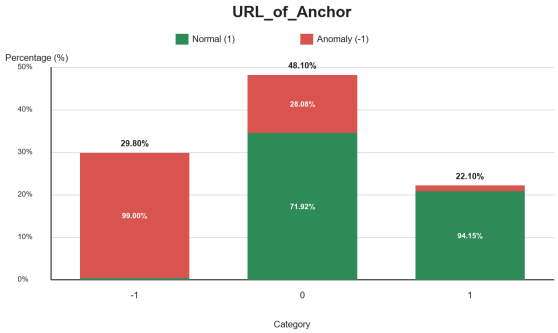


Figure 1: (continued) (c)–(d): bar heights are category shares of training records; labels within bars are within-category class shares.

6 Experimental Setup

All methods share partitions and seeds 0–4; seeds alter models, not data. Labels/IDs are excluded. Legitimate validation supports early stopping; labeled validation selects parameters/thresholds. AE: Adam (10^{-3}), batch 128, maximum 200 epochs, patience 10.

Validation grids cover AE hidden layers $(64, 32)/(128, 64)$ and latent widths $\{2, 4, 8, 16\}$; IF trees $\{100, 300\}$, samples $\{256, 1024\}$, feature fractions $\{.5, 1\}$; XGBoost trees $\{200, 500\}$, depths $\{3, 6, 9\}$, rates $\{.05, .1\}$. We use PyTorch 2.7.1, scikit-learn 1.6.1, XGBoost 3.0.1, and SciPy 1.15.2 on an AMD Ryzen 5 5600H CPU, no GPU.

Metrics: AP (average precision, not trapezoidal PR-AUC), ROC-AUC, F1, recall, and $FPR = FP/(FP+TN)$. We report five-seed means/SDs, paired seed-wise t-tests, and 2,000 paired site bootstraps of seed-averaged scores.

Overlap limitation: 987/1,659 test rows share training vectors (829 distinct); 672 are unseen. The split is not group-disjoint; unseen-vector results are sensitivity checks. Site bootstraps ignore vector-cluster dependence, limiting generalization claims.

7 Results

7.1 Main comparison

AE exceeds heuristic/IF but trails supervised XGBoost (Table 3; Figure 3). Versus IF, AP gap is $+.2919$ (95% bootstrap CI $[.2583, .3232]$, paired $p = 1.03 \times 10^{-6}$); versus XGBoost, $-.0498$ ($[-.0592, -.0401]$, $p = 1.78 \times 10^{-6}$). Gaps use

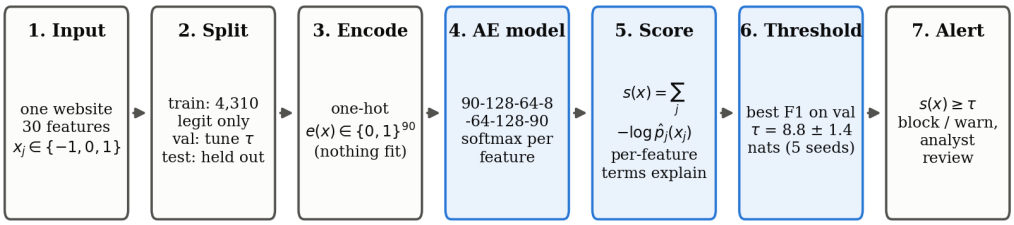


Figure 2: Our categorical AE pipeline. Threshold summary spans five seeds; we fit parameters on legitimate training records only.

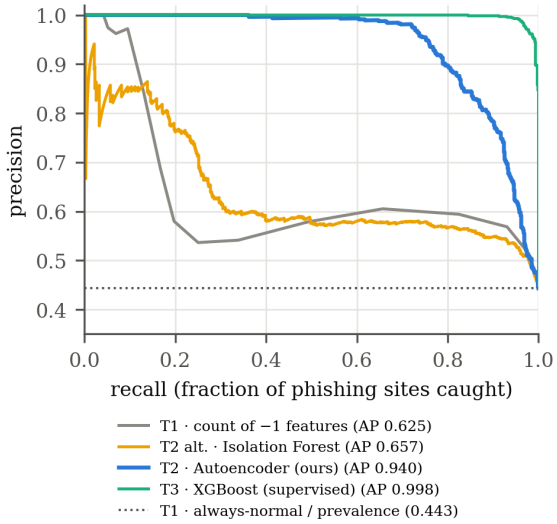


Figure 3: Our seed-0 test PR curves. The dotted line is a constant-score AP reference, not operational precision of a no-alert detector.

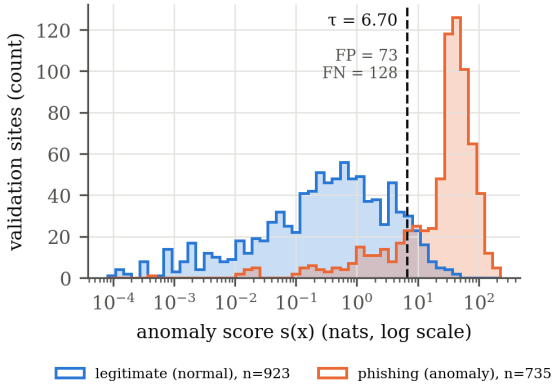


Figure 4: Our seed-0 validation scores and $\tau = 6.70$ nats. FP=73/FN=128 are validation counts; test counts are FP=92/FN=126.

AP of seed-averaged scores; Table 3 averages per-seed AP. Nonlinearity explains its different differences ($+.283/-.058$). Unseen-vector AP is .943 (AE) versus .998 (XGBoost), with prevalence .567 rather than .443; overlap is not thereby harmless. Figure 4 shows below-threshold phishing. AE takes 8.41 s to train and .002 ms/site in batch, excluding feature extraction: stronger ranking than IF at higher training cost, but weaker than supervised XGBoost.

7.2 Controlled ablations

Ablations hold split, seeds and validation-F1 selection fixed. Their baseline AP/F1 .939/.851 differs from main-run .940/.856 by .001/.005, within one seed SD (.003/.009); separate run sets do not imply identical conditions.

Top-9 AP is $.933 \pm .010$ (gap $-.0063$, $p = .242$), F1 .854: descriptive H1 support, not preregistered equivalence. Top-15 AP .951 means nine is not proven optimal (Table 4). H2 has partial support: zero Top-9 pairs at $V \geq .80$, one at .50 (Table 2). Close AP supports useful combined information, not conditional complementarity/minimality; leave-one-feature-out tests remain necessary.

With $n = 4,310$, 1/5/10% contamination lowers AP to .865/.779/.709; 10% raises FPR from 7.5% to 48.1%. CE training/MSE scoring gives AP .914, FPR 14.0%. These effects survive Bonferroni ($\alpha = .05/7 \approx .00714$), unlike Top-15 ($p = .0216$). Clean data and categorical scoring matter most.

8 Analysis

8.1 Error Analysis

At $\tau = 6.696$ nats, seed 0 yields 126 FNs/92 FPs on 1,659 test sites (TP=609, TN=832; Table 5). Near misses satisfy $.5 \leq s/\tau \leq 2$; inversions are at least tenfold off. Hamming distance uses the nearest legitimate training site. Conflicts suggest label noise or lossy representation; static codes have no sensor noise.

Table 6 selects the lowest-score conflicting FN, lowest-score systematic unseen-vector FN, highest-score FN, lowest/highest-score FP, and systematic FP with a legitimate training vector. Figure 5(a)–(c) shows five score orders among FNs, near-threshold FPs, and weak negative-

Tier	Method	AP	ROC-AUC	F1	FPR ↓	AP unseen	Train s / ms site
1	Always normal	.443±.000	.500±.000	.000±.000	.000±.000	.567±.000	0 / <.001
1	Count of −1	.625±.000	.710±.000	.706±.000	.562±.000	.690±.000	0 / <.001
2 alt.	Isolation Forest	.657±.014	.724±.013	.690±.010	.548±.020	.728±.009	.42 / .029
2	Autoencoder	.940±.003	.933±.003	.856±.009	.071±.017	.943±.004	8.41 / .002
3	XGBoost	**.998±.000**	**.998±.000**	**.973±.001**	.029±.001	**.998±.000**	.47 / .003

Table 3: Our test results, mean ± SD over five seeds; unseen-vector $n = 672$. AP denotes average precision. Latency excludes feature extraction; zero rounded SD is not exact equality.

Configuration	AP ± SD	F1	p vs. full
Full 30 / clean / CE	.939±.003	.851	–
Top 5	.932±.011	.862	.304
Top 9 (H1/H2)	.933±.010	.854	.242
Top 15	.951±.004	.873	.0216
Contamination 1%	.865±.011	.802	.000168
Contamination 5%	.779±.019	.748	.0000262
Contamination 10%	.709±.018	.693	.0000131
MSE scoring	.914±.003	.835	.000504

Table 4: Our controlled ablations across five seeds. Uncorrected paired t-test p values; Bonferroni criterion $\alpha = .05/7 \approx .00714$ for seven comparisons.

Property	FN ($n = 126$)	FP ($n = 92$)
Near miss (within 2×)	46 (37%)	61 (66%)
Complete inversion ($\geq 10\times$)	39 (31%)	0 (0%)
Median s/τ	.34	1.62
≤ 1 code from legit. training	71 (56%)	46 (50%)
Identical vector: both labels	17 (13%)	0 (0%)
Wrong in all five seeds	86 (68%)	13 (14%)
XGBoost correct	115 (91%)	86 (93%)

Table 5: Breakdown of all 218 seed-0 test errors.

code signals in missed phishing. First fixes are richer features/limited labels for FNs and FPR constraints/tested score caps for FPs; threshold changes cannot rescue near-zero-score phishing.

Case notes. (1) Training vector: five legitimate/four phishing; all $s_j \approx 0$, XGBoost $P = .50$. (2) Legitimate-like codes; all $s_j \approx 0$, $P = .03$. (3) SSL -1 contributes 3.60 nats; score is .38 below τ, $P = .73$. (4) Benign deviations add up: Traffic 0 (1.92), on_mouseover -1 (1.80), Abnormal_URL -1 (1.21). (5) Rare legitimate SSL -1 contributes 20.2 (46%); contextual mismatch does not establish temporal drift. (6) Vector occurs twice as legitimate in training; SFH $+1$ contributes 8.6 (63%), and all five seeds flag it. Feature terms sum to the total score.

8.2 Interpretability and faithfulness

For alert `test_00505` ($s = 40.2$, $\tau = 6.696$), anchor/SSL/tag-link terms are 26.0/7.4/5.6 nats (97%). Mean anchor/SSL shares are 35%/17%. AE rankings correlate with XGBoost mean absolute SHAP (Lundberg et al., 2020) ($\rho = .81$) and training MI (.83); agreement does not establish causation.

Resetting top 1/2/5 score-driving features to legitimate modes clears 30/54/70% of 609 true alerts, versus 2%/9% for one/five random features (20 repeats). This supports score sensitivity, but edited combinations may be implausible. Explanations support review of coded evidence, not raw pages or near-zero-score attacks.

8.3 Practical impact

At 1% phishing prevalence, 10,000 scans contain 100 phishing/9,900 legitimate sites (Table 7). Holding conditional error rates fixed, FPR constraints raise projected precision from .10 to .41 at 11 extra misses per 100 phishing sites; these are not deployment measurements.

9 Conclusion & Limitations

9.1 Key decisions, redesign, and future work

Clean normal-only training was our key decision: 10% contamination lowers AP from .939 to .709. CE scoring and validation-only thresholds are supported. H1 shows closeness, not equivalence; H2 remains cutoff-dependent.

If restarting, we would first freeze a vector-group-disjoint split, audit label conflicts, and specify an FPR budget and H1 equivalence margin. Richer features, limited labels, and score caps remain future tests.

9.2 Deployment limitations

Missing URLs/HTML/timestamps, conflicting labels, cross-split overlap, and site-bootstrap dependence limit generalization. The 44.3% prevalence differs from deployment; future distributions remain untested. AE needs clean,

#	Type	Site	SSL	A	T	L	$s(x)$	s/τ	Cause
1	FN	test_01368	+1	0	+1	-1	.0002	Inversion (3×10^{-5})	Label conflict
2	FN	test_01163	+1	0	+1	0	.0056	Inversion (8×10^{-4})	Missing information
3	FN	test_01618	-1	0	+1	+1	6.32	Near (.94)	Overlap
4	FP	test_01277	+1	+1	0	+1	6.73	Near (1.01)	Overlap
5	FP	test_01494	-1	0	+1	-1	44.3	Far (6.6)	Contextual mismatch
6	FP	test_00052	+1	+1	-1	0	13.6	Far (2.0)	Overlap
Legitimate training mode			+1	0	+1	0			
Phishing training mode			-1	-1	0	-1			

Table 6: Six cases, $\tau = 6.696$ nats. A=URL_of_Anchor; T=web_traffic; L=Links_in_tags; SSL=SSLfinal_State. Scores/ratios are rounded; case 6 is just above $2\times$. Signs have feature-specific meanings.

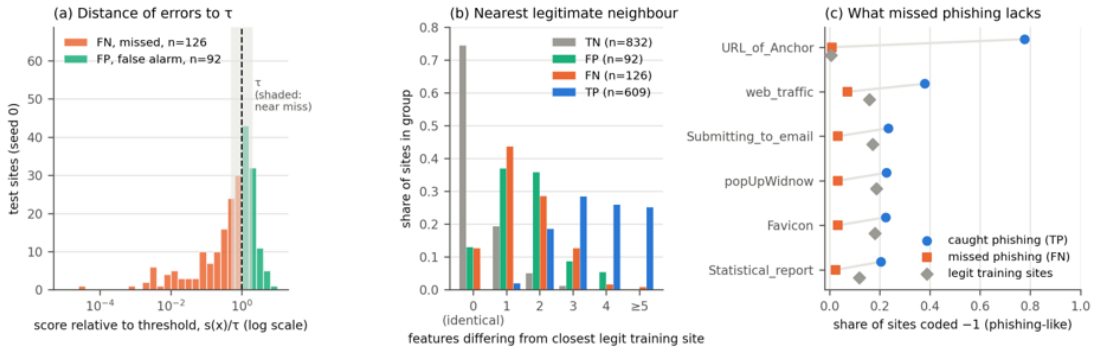


Figure 5: Seed-0 test error analysis: (a) scores relative to τ (log scale); (b) nearest legitimate-training Hamming distance by outcome; (c) negative-code shares for six features distinguishing caught from missed phishing.

AE threshold rule	Caught	Missed	False alerts
Validation best F1	81	19	701
Validation FPR $\leq$ 1%	70	30	99

Table 7: Our projected counts per 10,000 scans at 1% phishing prevalence, holding conditional error rates fixed.

representative normal data and drift monitoring/recalibration/retraining. Batch .002 ms/site excludes feature extraction and end-to-end delays. Deployment requires privacy-conscious processing and human review.

Contributions of Each Member

1 [515559007 馮昱祥]: Cleaned dataset, feature definitions (§3, Table 1); ACL formatting and GitHub upload.
2 [314551160 林暉庭]: Train/validation/test split, preprocessing pipeline, ground-truth labels (§3, Table 1); MI/redundancy, H1/H2 (§4, Table 2, Figure 1).
3 [315561001 黃少鋒]: Tier-2 categorical AE (one-hot, 8-unit bottleneck, normal-only training), CE score $s(x)$ with per-feature terms, validation-F1 and FPR$\leq$1% threshold rules (§5.1–5.2, Figure 2).
4 [315561002 莊誌強]: Tier-1 (always-normal, -1 count), Isolation Forest and XGBoost baselines, validation grid search, shared split, seeds 0–4, leakage rules and metrics (§5.3, §6).
5 [315554053 邱育宏]: Comparison/timing/significance (§7.1, Table 3, Figures 3–4).
6 [315551146 吳宥駒]: Ablations/hypotheses (§7.2, Table 4).
7 [315554001 杜洺鋒]: Errors/explanations/limitations (§§8–9, Tables 5–7, Figure 5).

AI usage. ChatGPT/Codex : wording, integration, audit,layout.

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